Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/16/2006
PI Name:	Martinez, Daniel A Ph.D.		
Project Title:	Healing Response of Injured Rodent Knee Ligaments to Hindlimb Unweighting and the Therapeutic Efficacy of Growth Hormone		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
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PI Email:	ddam@UH.edu	**Fax:**	FY 713-743-2636
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-2662
Organization Name:	Connective Tissue Physiology Laboratory University of Houston		
PI Address 1:	Department of Biology and Biochemistry		
PI Address 2:	369 Science and Research Building II		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-5001	**Congressional District:**	18
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:	Martinez, Daniel		
COI Name (Institution):	Vailas, Arthur C (University of Houston) Vanderby Jr., Ray (University of Wisconsin-Madison) Grindeland, Richard E (NASA-Ames Research Center)		
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Task Description:	Short-term and long-term exposure to either microgravity or hindlimb unloading has proven to suppress connective tissue formation in a variety of tissue types. In the growing rodent, hindlimb unloading causes a significant reduction in new bone formation and in knee ligament strength. Additionally, microgravity and hindlimb unloading also dramatically suppress immune function and endocrine function, specifically the reduction of growth hormone secretion in rodents, non-human primates, and humans. In essence, spaceflight has been shown to induce musculoskeletal weakness and reduce the release of various anabolic hormones responsible for growth promoting processes. Normal dense fibrous connective tissue regeneration is significantly impacted by tissue load histories and responds to growth promoting hormones and various cytokines. The rodent model for knee ligament injury and repair has been well characterized and has been used to study wound healing mechanisms by serving as a test physiological system for implementing countermeasures. The primary objective of our proposal is to use the rodent knee ligament injury repair model to study the impact hindlimb unloading has on the wound healing processes. We hypothesize that hindlimb unloading will (1) diminish the ability of knee ligaments to repair a surgical lesion, and (2) compromise the mechanical and biochemical properties of the ligaments, thus producing inferior material at the site of the ligament lesion. Growth hormone (GH) and insulin-like growth factor 1 (IGF-1) are potent stimulators of protein synthesis. We have given recombinant human GH (rhGH) and IGF-1 to promote wound healing during hindlimb unloading in the second half of our study. This study focuses on critical regenerative pathways of the injured knee ligaments after three weeks and seven weeks post-surgery. Collagen metabolism, tissue cytokine gene expression, collagen fibril organization, and tissue material properties in surgically oblated and non-surgical sham ligaments have been thoroughly examined. Based on our ground-based hindlimb unloading data, similar short term and long term spaceflight studies will be proposed testing new hypothesis and IGF-1 supplementation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will provide essential insight on injury response to microgravity. Clinicians can use the data from this study to determine the time course effects of unweighting on the healing response of knee ligaments. Information generated from this study will aid in the development of appropriate countermeasures for astronauts who acquire soft tissue injuries during chronic exposure to microgravity. Secondly, since GH secretion is compromised in animals and human astronauts during acute and chronic spaceflight, the assessment of the therapeutic efficacy of rhGH on soft tissue recovery as a hormonal countermeasure during hindlimb unloading has intrigue. rhGH and IGF-I will be investigated as a potential anabolic modulator to decrease the wound healing duration in a simulated microgravity environment. This therapy may prove useful not only in the spaceflight environment, but could be targeted to site specific tissues to decrease the time needed for regeneration and provide a rapid healing response during normal orthopaedic surgery.
Task Progress:	The surgeries and hindlimb unweighting experimental protocol has proven to be a successful animal model to study unloading on medial collateral ligament (MCL) wound healing. The Y01 and Y02 biomechanical, biochemical, and molecular analyses have been completed. We have presented the data on the time course of MCL wound healing events and hindlimb unweighting at four National Meetings in 2000 and 2001. Hindlimb unweighting (HLS) has a profound effect on ligament strength, matrix organization and the rate of ligament repair during wound healing. The integrity of the ligament is compromised during hindlimb unloading and the addition of time (3 weeks + an additional 4 weeks = Total 7 weeks) as a component to aid in healing actually further delays the wound repair process in the MCL. IGF-1 showed a significant effect in ultimate strength, stress and modulus in ambulatory animals, which was significantly less in the hindlimb unloaded animals. The combination of GH+IGF-1 revealed a significant effect in hindlimb unloaded animals, which was not present in ambulatory animals. The strength gains in the IGF-1 treated wounded ligaments can be attributed to increases in collagen gene expression (Collagen type I and III) mature collagen cross-link formation, collagen organization and decreased tissue cellularity. IGF-1 and GH+IGF-1 treated HLS+lesion groups similarly increased MCL tissue strength, collagen gene expression and tissue cellularity decreased. Together, these data suggest that IGF-1 alone or in a combination with rhGH may be a viable countermeasure to facilitate improved MCL wound healing. FUTURE INVESTIGATIONS DURING Y04 Further studies on the molecular gene expression of up- and down-regulated genes associated with wound healing and unloading are being studied using Micro Array gene chip technology. Two thousand five hundred genes and expressed sequence tags (ESTs) are currently being analyzed. MCL cDNA libraries will be constructed to screen potential regulatory genes and signaling pathways. Temporal and spatial distribution of collagen proteins and genes will be analyzed by immunohistochemistry and in situ mRNA screening on tissue sections from MCL samples not used in other analysis during the final year No Cost Extension.
Bibliography Type:	Description: (Last Updated: 04/23/2019)
Articles in Peer-reviewed Journals	Provenzano PP, Alejandro-Osorio AL, Grorud KW, Martinez DA, Vailas AC, Grindeland RE, Vanderby R Jr. "Systemic administration of IGF-I enhances healing in collagenous extracellular matrices: evaluation of loaded and unloaded ligaments." BMC Physiol. 2007 Mar 26;7:2. https://doi.org/10.1186/1472-6793-7-2 ; PubMed PMID: 17386107; PubMed Central PMCID: PMC1851714 , Mar-2007
Articles in Peer-reviewed Journals	Martinez DA, Vailas AC, Vanderby R Jr, Grindeland RE. "Temporal extracellular matrix adaptations in ligament during wound healing and hindlimb unloading." Am J Physiol Regul Integr Comp Physiol. 2007 Oct;293(4):R1552-60. Epub 2007 Aug 15. https://doi.org/10.1152/ajpregu.00423.2007 ; PubMed PMID: 17699562 , Oct-2007